

Physics 829: Homework Set No. 3

Due date: Friday, April 20, 2012, 1:00pm
in PRB M2043 (Biao Huang's office)

Total point value of set: 60 points

Problem 1 (10 pts.):

If $\hat{H}(\lambda)$ has eigenvector $|\psi_n(\lambda)\rangle$, prove the *Hellmann-Feynman Theorem*

$$\frac{dE_n(\lambda)}{d\lambda} = \left\langle \psi_n(\lambda) \left| \frac{\partial \hat{H}}{\partial \lambda} \right| \psi_n(\lambda) \right\rangle.$$

Problem 2 (15 pts.): Exercise 17.3.2 (Shankar, p. 466)

Problem 3 (15 pts.): Exercise 17.3.4 (Shankar, p. 470)

Problem 4 (5 pts.): Exercise 18.2.3 (Shankar, p. 478)

Problem 5 (10 pts.): Exercise 18.2.5 (Shankar, p. 478)

Problem 6 (5 pts.): Exercise 18.2.6 (Shankar, p. 478)

Problem 1

$$\frac{dE_n(\lambda)}{d\lambda} = \frac{d}{d\lambda} \langle \psi_n(\lambda) | E_n(\lambda) | \psi_n(\lambda) \rangle$$

$$\left(\text{since } \frac{d}{d\lambda} \underbrace{\langle \psi_n(\lambda) | \psi_n(\lambda) \rangle}_1 = 0 \right)$$

$$= \frac{d}{d\lambda} \langle \psi_n(\lambda) | \hat{H}(\lambda) | \psi_n(\lambda) \rangle$$

$$= \left(\frac{d}{d\lambda} \langle \psi_n(\lambda) | \right) \underbrace{\hat{H}(\lambda) | \psi_n(\lambda) \rangle}_{= E_n(\lambda) | \psi_n(\lambda) \rangle} + \langle \psi_n(\lambda) | \frac{\partial \hat{H}}{\partial \lambda} | \psi_n(\lambda) \rangle + \underbrace{\langle \psi_n(\lambda) | \hat{H}(\lambda) \left(\frac{d}{d\lambda} | \psi_n(\lambda) \rangle \right)}_{E_n(\lambda) \langle \psi_n(\lambda) |}$$

$$= \langle \psi_n(\lambda) | \frac{\partial \hat{H}}{\partial \lambda} | \psi_n(\lambda) \rangle + E_n(\lambda) \underbrace{\left(\left(\frac{d}{d\lambda} \langle \psi_n(\lambda) | \right) | \psi_n(\lambda) \rangle + \langle \psi_n(\lambda) | \left(\frac{d}{d\lambda} | \psi_n(\lambda) \rangle \right) \right)}_{= \frac{d}{d\lambda} \langle \psi_n(\lambda) | \psi_n(\lambda) \rangle = 0}$$

$$\Rightarrow \underline{\underline{\frac{dE_n(\lambda)}{d\lambda} = \langle \psi_n(\lambda) | \frac{\partial \hat{H}(\lambda)}{\partial \lambda} | \psi_n(\lambda) \rangle}} \quad \checkmark$$

Exercise 17.3.2 (p. 466)

Spin-1 matrices

$$S_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$S_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

$$S_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\left. \begin{aligned} S_x^2 &= \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix} \\ S_y^2 &= \frac{1}{2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{pmatrix} \end{aligned} \right\} S_x^2 - S_y^2 = \frac{1}{2} \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$S_z^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\hat{H} = \underbrace{A \hat{S}_z^2}_{\hat{H}^0} + \underbrace{B(S_x^2 - S_y^2)}_{\hat{H}'} \quad A \gg B$$

Use $\hat{S}_z$ -eigenbasis to represent $\hat{H}$:

$$\hat{H} \leftrightarrow \underbrace{\begin{pmatrix} A & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & A \end{pmatrix}}_{\hat{H}^0} + \underbrace{\begin{pmatrix} 0 & 0 & B \\ 0 & 0 & 0 \\ B & 0 & 0 \end{pmatrix}}_{\hat{H}'} = \begin{pmatrix} A & 0 & B \\ 0 & 0 & 0 \\ B & 0 & A \end{pmatrix}$$

• Exact eigenvalues and eigenstates of $\hat{H}$:

$$\det \begin{pmatrix} A-\lambda & 0 & B \\ 0 & -\lambda & 0 \\ B & 0 & A-\lambda \end{pmatrix} = -\det \begin{pmatrix} A-\lambda & B & 0 \\ 0 & 0 & -\lambda \\ B & A-\lambda & 0 \end{pmatrix} = +\det \begin{pmatrix} A-\lambda & B & 0 \\ B & A-\lambda & 0 \\ 0 & 0 & -\lambda \end{pmatrix}$$

$$= ((A-\lambda)^2 - B^2)(-\lambda) = 0$$

$$\Rightarrow \boxed{\lambda_1 = 0} \quad \text{or} \quad \begin{aligned} (\lambda - A)^2 &= B^2 \\ \lambda - A &= \pm B \end{aligned}$$

$$\boxed{\lambda_{2/3} = A \pm B}$$

$$\bullet \lambda_1 = 0: \begin{pmatrix} A & 0 & B \\ 0 & 0 & 0 \\ B & 0 & A \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = 0 \Rightarrow \begin{aligned} A\alpha + B\gamma &= 0 \\ \beta &= \text{arbitrary} \\ B\alpha + A\gamma &= 0 \end{aligned} \Rightarrow \alpha = -\frac{A}{B}\gamma \quad \left. \begin{aligned} A\alpha + B\gamma &= 0 \\ B\alpha + A\gamma &= 0 \end{aligned} \right\} (A^2 - B^2)\gamma = 0 \Rightarrow \gamma = \alpha = 0$$

$$\Rightarrow |1\rangle \leftrightarrow \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\underline{\lambda_2 = A+B}: \begin{pmatrix} -B & 0 & B \\ 0 & -A-B & 0 \\ B & 0 & -B \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} B(-\alpha + \gamma) = 0 \\ (A+B)\beta = 0 \Rightarrow \beta = 0 \\ B(\alpha - \gamma) = 0 \end{cases}$$

$$\Rightarrow \gamma = \alpha$$

$$\Rightarrow |2\rangle \leftrightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$\underline{\lambda_3 = A-B}: \begin{pmatrix} B & 0 & B \\ 0 & -A+B & 0 \\ B & 0 & B \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} B(\alpha + \gamma) = 0 \\ (-A+B)\beta = 0 \Rightarrow \beta = 0 \\ B(\alpha + \gamma) = 0 \end{cases}$$

$$\Rightarrow \gamma = -\alpha$$

$$\Rightarrow |3\rangle \leftrightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

• Now the perturbative approach.

- $\hat{H}^0$ has 2 eigenvalues, 0 and 1, with 1 being twofold degenerate.

$$\underline{\lambda_1^0 = 0}, \quad \underline{\lambda_2^0 = \lambda_3^0 = 1}$$

- The corresponding eigenstates are

$$\begin{cases} |1^0\rangle \leftrightarrow \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \\ |2^0\rangle \leftrightarrow \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\ |3^0\rangle \leftrightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \end{cases}$$

The eigenstate $|1^0\rangle$ is stable under the perturbation:

$$\langle m^0 | \hat{H}' | 1^0 \rangle = (\alpha^* \beta^* \gamma^*) \begin{pmatrix} 0 & 0 & B \\ 0 & 0 & 0 \\ B & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0 \quad \text{for all } \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$$

$\Rightarrow \hat{H}'$ has no matrix elements mixing the state $|1^0\rangle$ with the other eigenstates of $\hat{H}^0$.

- To see the eigenstates in the degenerate subspace $\{|2^0\rangle, |3^0\rangle\}$ that are stable under the perturbation, we must diagonalize the truncated Hamiltonian

$$\hat{H} = \hat{P} \hat{H} \hat{P} = \hat{P} (\hat{H}^0 + \hat{H}') \hat{P} \leftrightarrow \begin{pmatrix} A & B \\ B & A \end{pmatrix} \begin{matrix} \leftarrow |2^0\rangle \\ \leftarrow |3^0\rangle \end{matrix}$$

$\begin{matrix} |2^0\rangle & |3^0\rangle \\ \uparrow & \downarrow \\ \langle 3^0| & \langle 2^0| \end{matrix}$

in this degenerate subspace.

To find the eigenvalues $\bar{\lambda}$ of $\hat{H}$ we compute

$$\det(\hat{H} - \bar{\lambda} \hat{1}) = 0 \Rightarrow \begin{vmatrix} A - \bar{\lambda} & B \\ B & A - \bar{\lambda} \end{vmatrix} = 0 \Rightarrow (A - \bar{\lambda})^2 - B^2 = 0$$

$$\Rightarrow (\bar{\lambda} - A)^2 = B^2 \Rightarrow \boxed{\bar{\lambda}_{2,3} = A \pm B}$$

These are the exact eigenvalues $\lambda_{2,3}$ of $\hat{H}$!

The corresponding eigenstates are (in the $\{|2^0\rangle, |3^0\rangle\}$ subspace)

$$\begin{pmatrix} \langle 2^0 | \bar{2} \rangle \\ \langle 3^0 | \bar{2} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} ; \quad \begin{pmatrix} \langle 2^0 | \bar{3} \rangle \\ \langle 3^0 | \bar{3} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

(using work from the exact solution)

$$\Rightarrow |\bar{2}\rangle = |2\rangle, \quad |\bar{3}\rangle = |3\rangle$$

The eigenstates that diagonalize $\hat{H}$ in the degenerate subspace corresponding to $\lambda_{2,3}^0 = 1$ are the exact eigenstates of $\hat{H}$, with the exact eigenvalues $\bar{\lambda}_{2,3} = \lambda_{2,3}$. This is so since $\hat{H}$ has zero matrix elements between the degenerate subspace $\{|2^0\rangle, |3^0\rangle\} = \{|2\rangle, |3\rangle\}$ and the rest of its Hilbert space (i.e. the 1-d subspace spanned by $|1^0\rangle = |1\rangle$).

Exercise 17.3.4 (p. 470)

(1) Consider $\hat{H}(\lambda) = \hat{H}_0 + \lambda \hat{H}' = \hat{H}_{\text{Coulomb}}(e^2) + \frac{\lambda}{r} = \hat{H}_{\text{Coulomb}}(e^2 - \lambda)$
 where $\hat{H}_{\text{Coulomb}}(e^2) = \frac{\hat{p}^2}{2m} - \frac{e^2}{r}$

We want $\langle \frac{1}{r} \rangle_{n\ell m} = \langle \psi_{n\ell m} | \hat{H}' | \psi_{n\ell m} \rangle$

We can compute the eigenenergies of $\hat{H}(\lambda)$ exactly; they are

$$E_n(\lambda) = -\frac{m(e^2 - \lambda)^2}{2\hbar^2 n^2} = -\frac{me^4}{2\hbar^2 n^2} + \frac{me^2 \lambda}{\hbar^2 n^2} - \frac{m\lambda^2}{2\hbar^2 n^2}$$

This contains corrections of order λ and λ^2 to $E_n(0) = -\frac{me^4}{2\hbar^2 n^2}$

The term $\sim \lambda$ can be identified with the first order correction

$$\lambda E_n^{(1)} = \langle \psi_{n\ell m}(\lambda=0) | \frac{\lambda}{r} | \psi_{n\ell m}(\lambda=0) \rangle$$

$$\Rightarrow \langle \frac{1}{r} \rangle_{n\ell m} = +\frac{me^2}{\hbar^2 n^2} = \frac{1}{n^2 a_0}$$

(2) Now consider $\hat{H}(\lambda) = \hat{H}_0 + \lambda \tilde{H}' = \hat{H}_{\text{Coulomb}} + \frac{\lambda}{r^2}$

In the coordinate representation, the term $\lambda \tilde{H}' = \frac{\lambda}{r^2}$ has the same r -dependence as the centrifugal term $\frac{\hbar^2}{2m} \frac{\ell(\ell+1)}{r^2}$. So the exact eigenfunctions of $\hat{H}(\lambda)$ are hydrogen eigenfunctions whose radial wavefunction are obtained from a modified radial differential equation where

$$\frac{\ell(\ell+1)}{r^2} \longrightarrow \frac{\tilde{\ell}(\tilde{\ell}+1)}{r^2} = \frac{\ell(\ell+1)}{r^2} + \frac{2m\lambda}{\hbar^2 r^2}$$

$$\text{or } \tilde{\ell}(\tilde{\ell}+1) = \ell(\ell+1) + \frac{2m\lambda}{\hbar^2} \Rightarrow (2\tilde{\ell}+1)d\tilde{\ell} = \frac{2m}{\hbar^2} d\lambda$$

l enters the energy eigenvalue through the main quantum number $n = k + l + 1$ where k is the number of nodes of the radial wavefunction:

$$\tilde{E}_{n\tilde{l}m}(\lambda) = -\frac{me^4}{2\hbar^2(k+\tilde{l}(\lambda)+1)^2} = \langle n\tilde{l}m; \lambda | \hat{H}(\lambda) | n\tilde{l}m; \lambda \rangle$$

The Hellmann-Feynman theorem gives

$$\begin{aligned} \frac{d\tilde{E}_{n\tilde{l}m}(\lambda)}{d\lambda} &= \langle n\tilde{l}m; \lambda | \frac{\partial \hat{H}(\lambda)}{\partial \lambda} | n\tilde{l}m; \lambda \rangle = \\ &= \langle n\tilde{l}m; \lambda | \frac{1}{r^2} | n\tilde{l}m; \lambda \rangle \end{aligned}$$

So we get $\langle \frac{1}{r^2} \rangle_{n\tilde{l}m}$ by evaluating this at $\lambda=0$:

$$\begin{aligned} \underbrace{\left\langle \frac{1}{r^2} \right\rangle_{n\tilde{l}m}} &= \left. \frac{dE_{n\tilde{l}m}}{d\lambda} \right|_{\lambda=0} = + \frac{me^4}{\hbar^2(k+\tilde{l}+1)^3} \left. \frac{d\tilde{l}}{d\lambda} \right|_{\lambda=0} \\ &= \frac{me^4}{\hbar^2(k+l+1)^3} \frac{2m}{\hbar^2} \frac{1}{2l+1} = \frac{2m^2e^4}{\hbar^4n^3} \frac{1}{2l+1} \\ &= \frac{2}{a_0^2n^3} \frac{1}{2l+1} = \underbrace{\frac{1}{l+1/2} \frac{1}{a_0^2n^3}} \end{aligned}$$

To compare with (17.3.11) we multiply with e^4 :

$$\underbrace{\left\langle \frac{e^4}{r^2} \right\rangle_{n\tilde{l}m}} = \frac{4m^2e^8}{4\hbar^4n^4} \cdot \frac{n}{l+1/2} = \frac{4n}{l+1/2} \left(-\frac{me^4}{2\hbar^2n^2} \right)^2 = \underbrace{\frac{4nE_0^2}{l+1/2}} \quad \checkmark$$

(3) Consider $\langle \frac{1}{r^3} \rangle_{n\tilde{l}m}$. The radial part of the kinetic energy operator in coordinate representation reads

$$-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r}$$

Define $\hat{p}_r \leftrightarrow -i\hbar \left(\frac{\partial}{\partial r} + \frac{1}{r} \right) = -i\hbar \left(\frac{1}{r} \frac{\partial}{\partial r} r \right)$

which is shorthand for

$$\langle r | \hat{p}_r | \psi \rangle = -i\hbar \frac{1}{r} \frac{\partial}{\partial r} (r \langle r | \psi \rangle) = -i\hbar \frac{1}{r} \frac{\partial}{\partial r} (r \psi(r))$$

Then

$$\begin{aligned} \langle r | \frac{\hat{p}_r^2}{2m} | \psi \rangle &= -\frac{\hbar^2}{2m} \frac{1}{r} \frac{\partial}{\partial r} r \left(\frac{1}{r} \frac{\partial}{\partial r} (r \psi(r)) \right) = -\frac{\hbar^2}{2m} \frac{1}{r} \frac{\partial^2}{\partial r^2} (r \psi(r)) \\ &= -\frac{\hbar^2}{2m} \frac{1}{r} \frac{\partial}{\partial r} (r \psi' + \psi) = -\frac{\hbar^2}{2m} \left(\psi'' + \frac{2}{r} \psi' \right) \end{aligned}$$

while

$$-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial \psi}{\partial r} = -\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \psi') = -\frac{\hbar^2}{2m} \left(\psi'' + \frac{2}{r} \psi' \right)$$

So the two are the same expressions.

Hence $\frac{\hat{p}_r^2}{2m}$ is the radial kinetic energy operator.

In an energy eigenstate

$$\begin{aligned} \langle nlm | [\hat{H}, \hat{p}_r] | nlm \rangle &= \langle nlm | \hat{H} \hat{p}_r | nlm \rangle - \langle nlm | \hat{p}_r \hat{H} | nlm \rangle \\ &= E_n \langle nlm | - E_n \langle nlm | \\ &= \langle nlm | \hat{p}_r | nlm \rangle (E_n - E_n) = 0 \end{aligned}$$

On the other hand,

$$\begin{aligned} [A, \hat{p}_r] &= \left[\frac{\hat{p}_r^2}{2m} + \frac{1}{2mr^2} \hat{L}^2 - \frac{e^2}{\hat{r}}, \hat{p}_r \right] = \\ &= \left[\hat{p}_r, \frac{e^2}{\hat{r}} - \frac{\hat{L}^2}{2m\hat{r}^2} \right] \quad (\hat{L}^2 \text{ and } \frac{1}{\hat{r}^2} \text{ commute}) \end{aligned}$$

In the r basis,

$$e^2 \left[\hat{p}_r, \frac{1}{\hat{r}} \right] = -\frac{e^2}{\hat{r}^2} \hat{p}_r$$

In the r basis,

$$\text{The commutator } \left[\hat{p}_r, \frac{1}{\hat{r}^2} \right] = \frac{1}{2m} \left[\hat{p}_r, \frac{1}{\hat{r}^2} \right] = \frac{\hbar^2}{2m} \left[\hat{p}_r, \frac{1}{\hat{r}^2} \right] = 0$$

$$\Rightarrow 0 = \langle nlm | [\hat{H}, \hat{p}_r] | nlm \rangle = \int d^3r \psi_{nlm}^*(\vec{r}) (-i\hbar) \left(\frac{\partial}{\partial r} + \frac{1}{r} \right) \left(\frac{e^2}{r} - \frac{\hbar^2 l(l+1)}{2mr^2} \right) \psi_{nlm}(\vec{r}) \\ - \int d^3r \psi_{nlm}^*(\vec{r}) \left(\frac{e^2}{r} - \frac{\hbar^2 l(l+1)}{2mr^2} \right) (-i\hbar) \left(\frac{\partial}{\partial r} + \frac{1}{r} \right) \psi_{nlm}(\vec{r})$$

(we used that $\psi_{nlm}(\vec{r}) = R_{nl}(r) Y_{lm}(\vartheta, \varphi)$)

$$\text{and } \hat{L}^2 Y_{lm}(\vartheta, \varphi) = \hbar^2 l(l+1) Y_{lm}(\vartheta, \varphi)$$

$$= -i\hbar \int d^3r \psi_{nlm}^*(\vec{r}) \left(-\frac{e^2}{r^2} + \frac{\hbar^2 l(l+1)}{mr^3} \right) \psi_{nlm}(\vec{r})$$

(all other terms cancel)

$$= -i\hbar \left[\left\langle -\frac{e^2}{r^2} \right\rangle_{nlm} + \frac{\hbar^2 l(l+1)}{m} \left\langle \frac{1}{r^3} \right\rangle_{nlm} \right]$$

$$\Rightarrow \left\langle \frac{1}{r^3} \right\rangle_{nlm} = \frac{me^2}{\hbar^2} \frac{1}{l(l+1)} \left\langle \frac{1}{r^2} \right\rangle_{nlm} = \frac{1}{l(l+1) a_0} \left\langle \frac{1}{r^2} \right\rangle_{nlm}$$

Plugging in the result from (2) this becomes

$$\left\langle \frac{1}{r^3} \right\rangle_{nlm} = \frac{1}{a_0^3 n^3} \frac{1}{l(l+\frac{1}{2})(l+1)} \quad (17.3.20)$$

$$(4) \text{ Consider } \hat{H}(\lambda) = \hat{H}_{\text{Coulomb}} + \lambda \frac{\hat{p}^2}{2m} \equiv \hat{H}_0 + \lambda \hat{H}'$$

$$\text{We want } \langle H' \rangle_{nlm} = \left\langle \frac{\hat{p}^2}{2m} \right\rangle_{nlm}$$

We compute the eigenvalues of $\hat{H}(\lambda)$ exactly:

$$\hat{H}(\lambda) = \hat{H}_{\text{Coulomb}} \left(\mu = \frac{m}{1+\lambda} \right)$$

$$\Rightarrow E_n(\lambda) = - \frac{me^4}{2\hbar^2(1+\lambda)n^2} = E_n^0 + \lambda \frac{dE_n(\lambda)}{d\lambda} \Big|_{\lambda=0} + \dots$$

$$= E_n^0 + \lambda E_n' + \dots$$

$$\Rightarrow \underline{E_n^{(1)}} = \langle H' \rangle_{nlm} = \frac{dE_n(\lambda)}{d\lambda} \Big|_{\lambda=0} = + \frac{me^4}{2\hbar^2(1+\lambda)^2 n^2} \Big|_{\lambda=0}$$

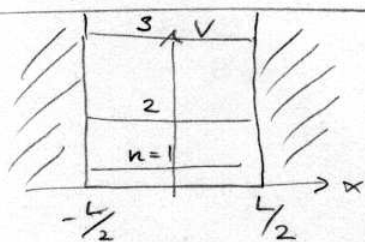
$$= \frac{me^4}{2\hbar^2 n^2} = -E_n^0 = -\underline{\langle H^0 \rangle_{nlm}}$$

$$\Rightarrow \langle \hat{T} \rangle_{nlm} = -\langle \hat{T} + \hat{V} \rangle_{nlm}$$

$$\Rightarrow \boxed{\langle T \rangle_{nlm} = -\frac{1}{2} \langle \hat{V} \rangle_{nlm}}$$

Virial theorem for hydrogen
(Coulomb potential) (13.1.33)

Exercise 18.2.3 (p. 478)



The semiclassical argument for the natural time period for a particle in the ground state of their box is given in the text on p. 479, Eq. (18.2.19)

Let's assume the box doubles its size in time $\tau \ll T \approx \frac{mL^2}{\pi\hbar}$.

This is a rapid change, and we can use the sudden approximation that $|\psi_{\text{after}}\rangle = |\psi_{\text{before}}\rangle = |\psi_{n=1,L}\rangle$ right after the change. The probability of catching the particle in the ground state of the new box is

$$\begin{aligned}
 \underbrace{P_{n=1}(2L)} &= |\langle n=1, 2L | n=1, L \rangle|^2 \\
 &= \left| \int_{-\infty}^{\infty} dx \sqrt{\frac{2}{2L}} \cos\left(\frac{\pi x}{2L}\right) \theta(L-|x|) \cdot \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{L}\right) \theta(L-|x|) \right|^2 \\
 &= \frac{2}{L^2} \left| \int_{-L/2}^{L/2} dx \cos\left(\frac{\pi x}{L}\right) \cos\left(\frac{\pi x}{2L}\right) \right|^2 \quad \xi = \frac{\pi x}{L} \\
 &= 2 \left| \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} d\xi \cos(\xi) \cos(\xi/2) \right|^2 \\
 &= 2 \left| \frac{1}{2\pi} \int_0^{\pi/2} d\xi \underbrace{(e^{i\xi} + e^{-i\xi})(e^{i\xi/2} + e^{-i\xi/2})}_{e^{3i\xi/2} + e^{i\xi/2} + e^{-i\xi/2} + e^{-3i\xi/2}} \right|^2 \\
 &= 2 \left| \frac{1}{\pi} \int_0^{\pi/2} d\xi (\cos(3\xi/2) + \cos(\xi/2)) \right|^2 \\
 &= \frac{2}{\pi^2} \left| \frac{2}{3} \sin\left(\frac{3\pi}{4}\right) + 2 \sin\left(\frac{\pi}{4}\right) \right|^2 = \left(\frac{8}{3\pi}\right)^2 \approx 0.721 = 72.1\%
 \end{aligned}$$

Exercise 18.2.5 (p.478)

$$\hat{H} = \hat{H}^0 + \hat{H}' = \frac{\hat{p}^2}{2m} + \frac{m}{2} \omega^2 \hat{x}^2 - f x \Theta(-t)$$

The external force f is abruptly turned off at $t=0$, and we can therefore use the sudden approximation that at $t=0^+$ the particle is still in the ground state of

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m}{2} \omega^2 \left(\hat{x} - \frac{f}{m\omega^2} \right)^2 - \frac{1}{2} \frac{f^2}{m\omega^2}$$

Its energy is $E_0 = \hbar\omega/2 - \frac{f^2}{2m\omega^2}$, and its wavefunction is

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-\frac{m\omega}{2\hbar} \left(x - \frac{f}{m\omega^2} \right)^2}$$

The corresponding state vector is

$$|0\rangle = \hat{T}\left(\frac{f}{m\omega^2}\right)|0^0\rangle = e^{-\frac{i}{\hbar} \frac{f}{m\omega^2} \hat{p}} |0^0\rangle \quad (17.2.9)$$

where $|0^0\rangle$ is the ground state of $\hat{H}^0 = \frac{\hat{p}^2}{2m} + \frac{m}{2} \omega^2 \hat{x}^2$.

We want

$$\begin{aligned} P(n) &= |\langle n^0 | 0 \rangle|^2 = |\langle n^0 | e^{-\frac{i}{\hbar} \frac{f}{m\omega^2} \hat{p}} | 0^0 \rangle|^2 \\ &= |\langle n^0 | e^{-\frac{i}{\hbar} \frac{f}{m\omega^2} \sqrt{\frac{\hbar m \omega}{2}} \frac{\hat{a} - \hat{a}^\dagger}{i}} | 0^0 \rangle|^2 \\ &= |\langle n^0 | e^{\frac{f}{\sqrt{2} \hbar m \omega^3} (\hat{a}^\dagger - \hat{a})} | 0^0 \rangle|^2 = |\langle n^0 | e^{\sqrt{\lambda} (\hat{a}^\dagger - \hat{a})} | 0^0 \rangle|^2 \end{aligned}$$

where $\lambda = \frac{f^2}{2\hbar m \omega^3}$.

$$\begin{aligned} \text{Now } e^{\sqrt{\lambda} (\hat{a}^\dagger - \hat{a})} &= e^{\sqrt{\lambda} \hat{a}^\dagger} e^{-\sqrt{\lambda} \hat{a}} e^{-\frac{1}{2} [\sqrt{\lambda} \hat{a}^\dagger, -\sqrt{\lambda} \hat{a}]} \quad [\hat{a}^\dagger, \hat{a}] = -1 \\ &= e^{-\lambda/2} e^{\sqrt{\lambda} \hat{a}^\dagger} e^{-\sqrt{\lambda} \hat{a}} \end{aligned}$$

and $e^{-\sqrt{\lambda}\hat{a}}|0^0\rangle = |0^0\rangle$

$$\Rightarrow P(n) = e^{-\lambda} |\langle n^0 | e^{\sqrt{\lambda}\hat{a}^\dagger} | 0^0 \rangle|^2 = e^{-\lambda} |\langle n^0 | \sum_{m=0}^{\infty} \frac{\lambda^{m/2}}{m!} (\hat{a}^\dagger)^m | 0^0 \rangle|^2$$

Now $|m^0\rangle = \frac{(\hat{a}^\dagger)^m}{\sqrt{m!}} |0^0\rangle$ and $\langle n^0 | m^0 \rangle = \delta_{mn}$

$$\begin{aligned} P(n) &= e^{-\lambda} \left| \langle n^0 | \sum_{m=0}^{\infty} \frac{\lambda^{m/2}}{\sqrt{m!}} | m^0 \rangle \right|^2 \\ &= e^{-\lambda} \left| \sum_{m=0}^{\infty} \frac{\lambda^{m/2}}{\sqrt{m!}} \underbrace{\langle n^0 | m^0 \rangle}_{\delta_{mn}} \right|^2 = e^{-\lambda} \left| \frac{\lambda^{n/2}}{\sqrt{n!}} \right|^2 \end{aligned}$$

$$\Rightarrow \underline{P(n) = \frac{e^{-\lambda} \lambda^n}{n!}} \quad \text{as desired}$$

Exercise 18.2.6 (p. 478)

$$\int_{t=0^-}^{t=0^+} i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle = (\hat{H}^0 + \hat{H}'\delta(t)) |\psi(t)\rangle$$

$$|\psi(0^+)\rangle - |\psi(0^-)\rangle = -\frac{i}{\hbar} \lim_{\epsilon \rightarrow 0} \int_{-\epsilon/2}^{\epsilon/2} (\hat{H}^0 + \hat{H}'\delta(t)) |\psi(t)\rangle dt$$

$$= -\frac{i}{\hbar} \hat{H}' |\psi(0)\rangle \quad (*)$$

If at $t=0^-$ $|\psi(0^-)\rangle = |i^0\rangle$, and we expand $|\psi(0^+)\rangle$ in eigenstates of $\hat{H}^0$,

$$|\psi(0^+)\rangle = \sum_n c_n |n^0\rangle,$$

then (using the fact that $|\psi(0)\rangle = |\psi(0^-)\rangle + O(\hat{H}') = |i^0\rangle + O(\hat{H}')$)

$$c_f = \langle f^0 | \psi(0^+) \rangle = \underbrace{\langle f^0 | i^0 \rangle}_{=0 \text{ since } f \neq i} - \frac{i}{\hbar} \langle f^0 | \hat{H}' | i^0 \rangle + \text{higher orders}$$

$$\Rightarrow \boxed{c_f^{(1)} = -\frac{i}{\hbar} \langle f^0 | \hat{H}' | i^0 \rangle}$$

This only good to first order in $\hat{H}'$, since we made the replacement $|\psi(0)\rangle \rightarrow |i^0\rangle$ on the r.h.s. of (*).